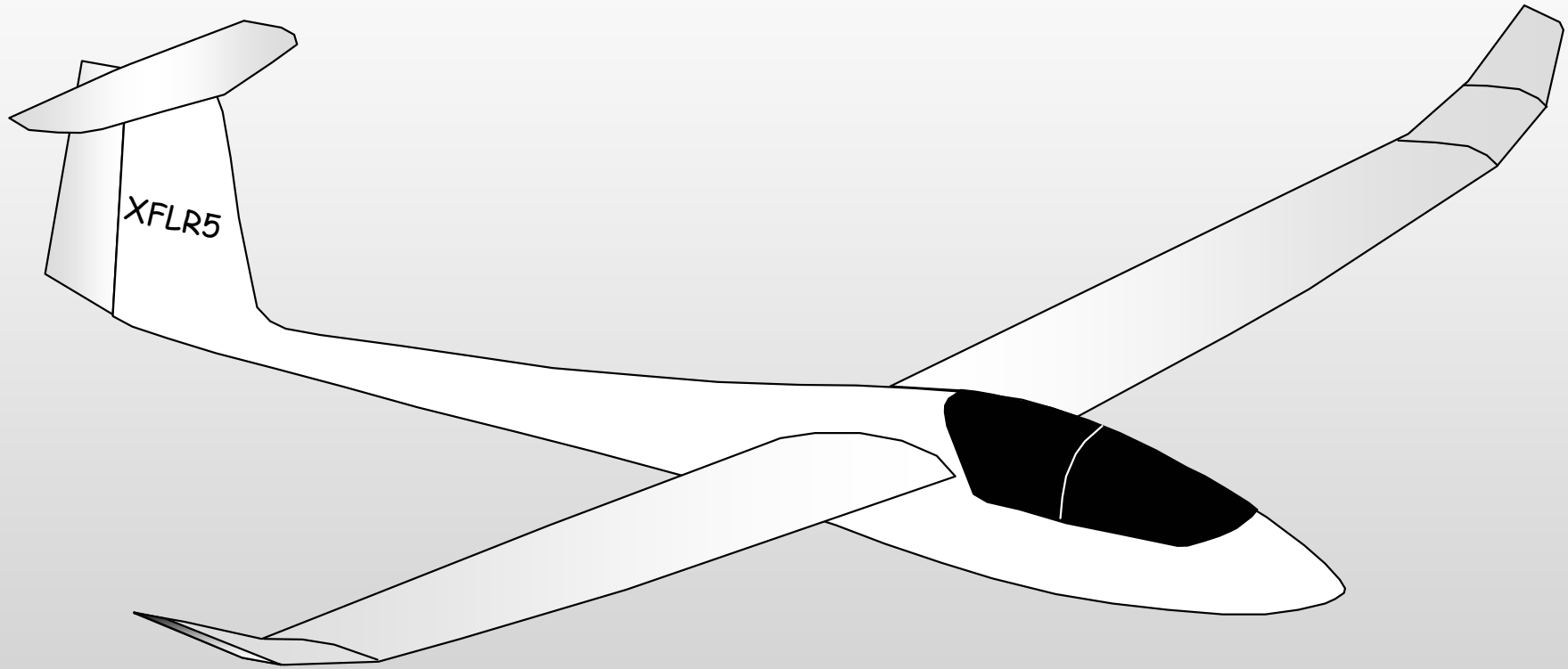
About performance and stability analysis using XFLR5



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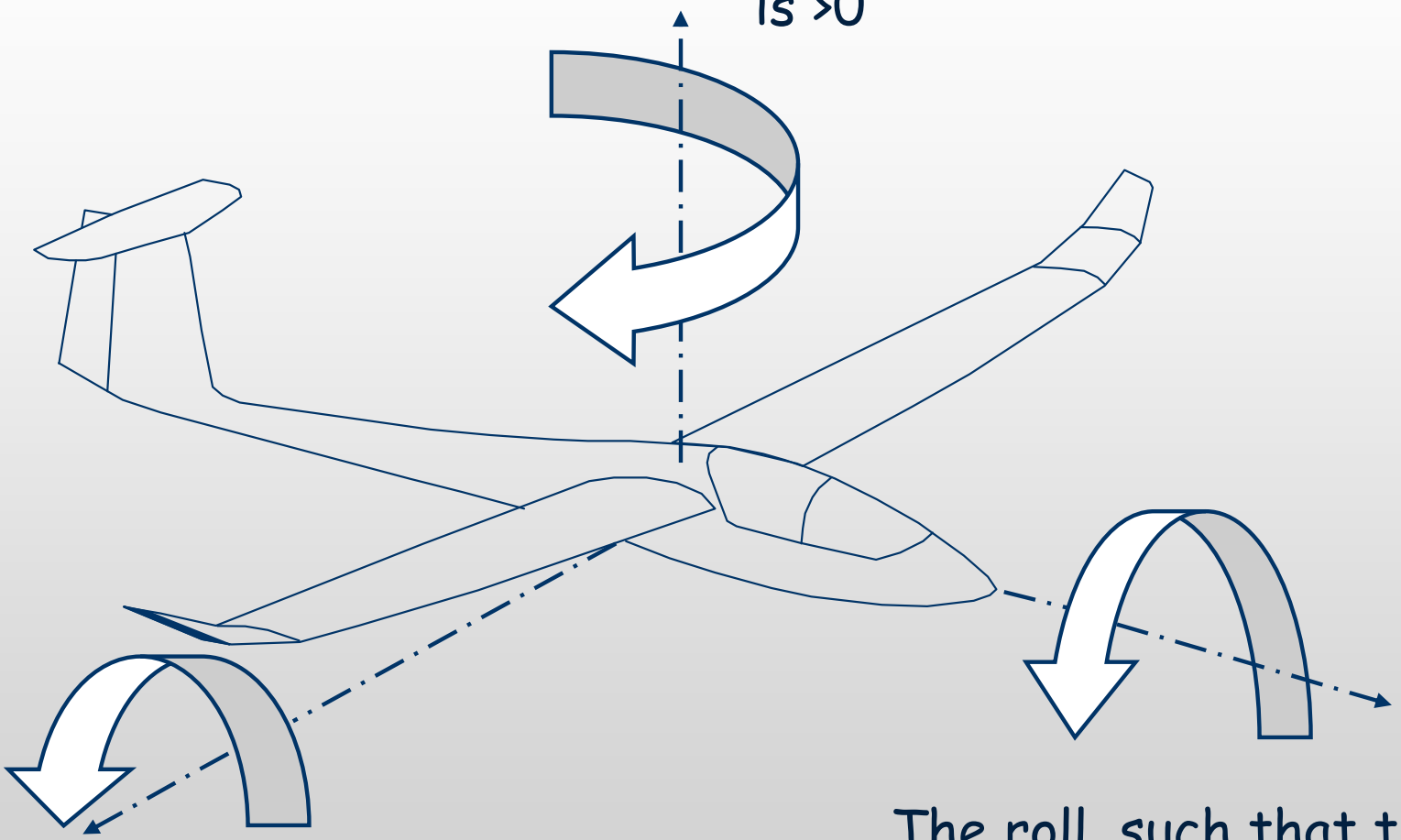
Sign Conventions



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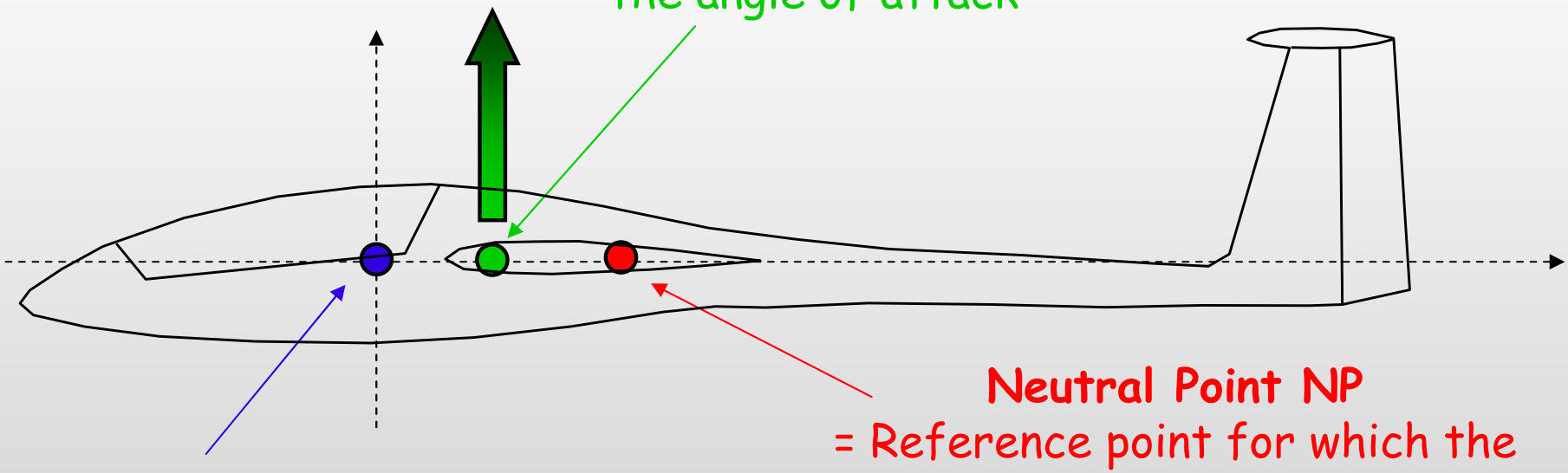
The three key points which must not be confused together

Centre of Pressure CP

= Point where the resulting aero force applies
Depends on the model's aerodynamics and on the angle of attack



Neutral Point NP

= Reference point for which the pitching moment does not depend on the angle of attack α

Depends only on the plane's external geometry

Not exactly intuitive, so let's explore the concept further

Centre of Gravity CG

= Point where the moments act;
Depends only on the plane's mass distribution, not its aerodynamics

Also named XCmRef in XFLR5, since this is the point about which the pitching moment is calculated

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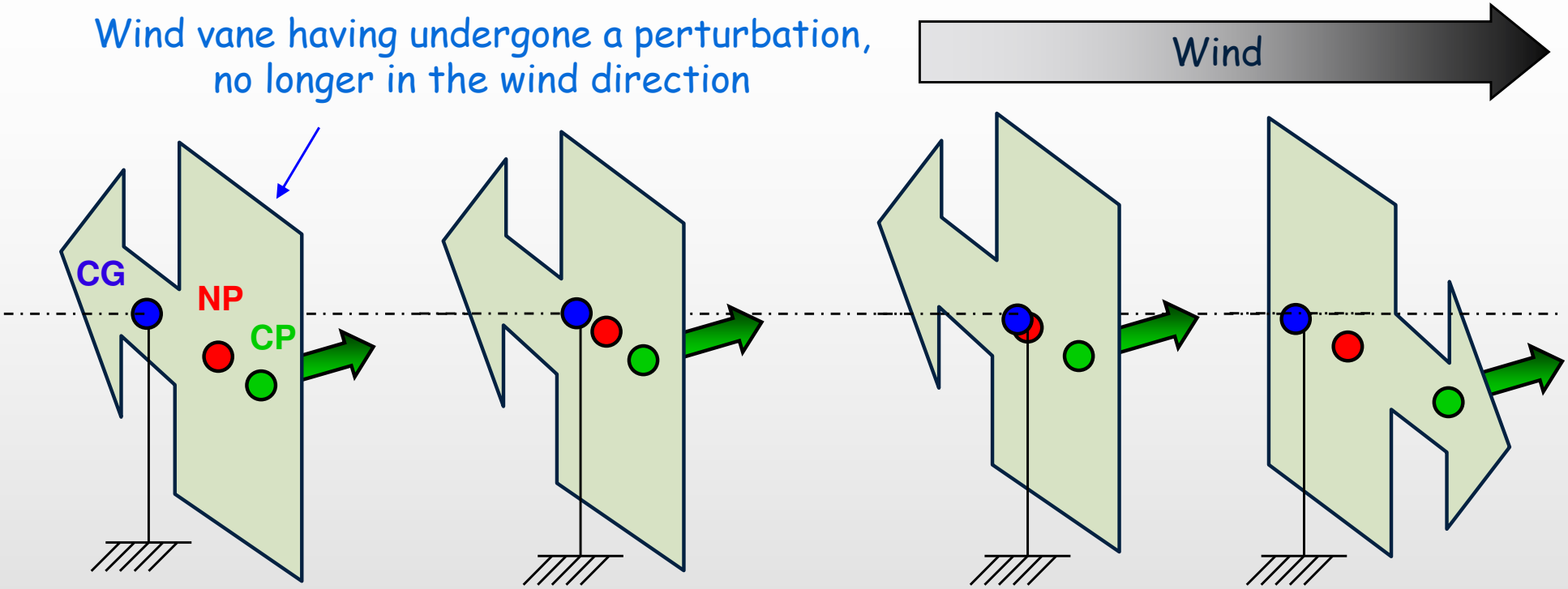
The neutral point = Analogy with the wind vane



CG forward of the NP
- → The pressure forces drive the vane back in the wind direction
- → Very stable wind vane

CG slightly forward of the NP
- → The pressure forces drive the vane back in the wind direction
- → The wind vane is stable, but sensitive to wind gusts

CG positioned at the NP
- → The wind vane rotates indefinitely
- → Unstable

CG behind the NP
- → The wind vane is stable… in the wrong direction

The Neutral Point is the rear limit for the CG

☞ **2nd principle : Forward of the NP, the CG thou shall position**

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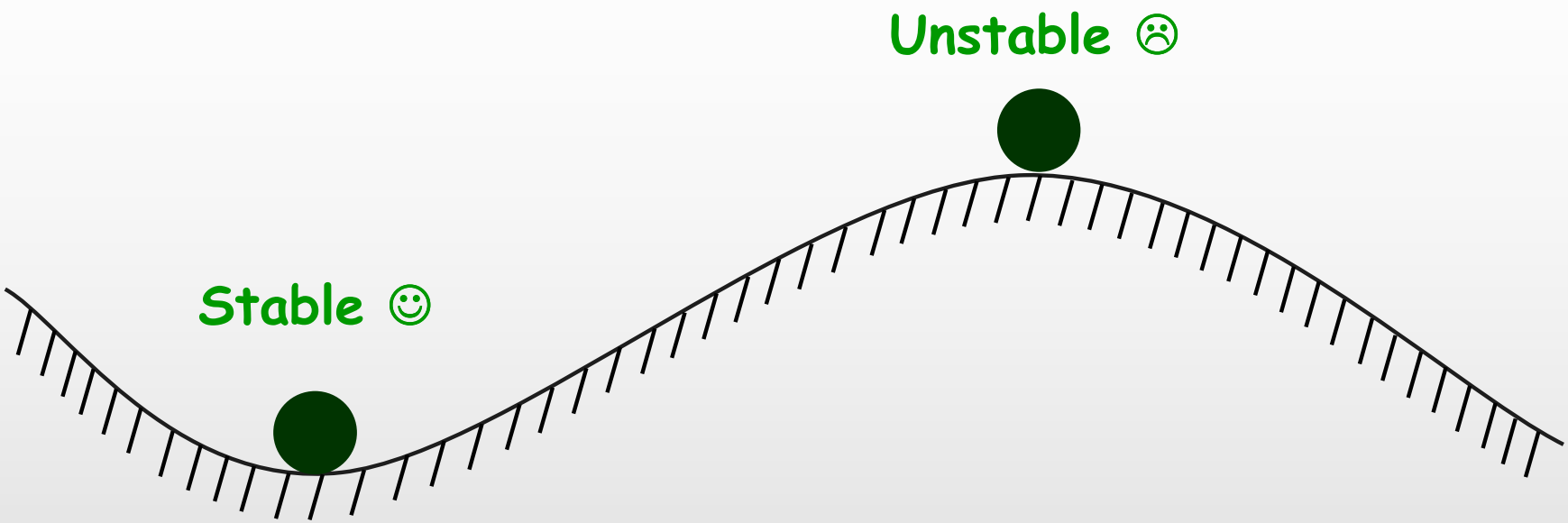
A preliminary note : Equilibrium is not stability !



Both positions are at equilibrium, only one is stable

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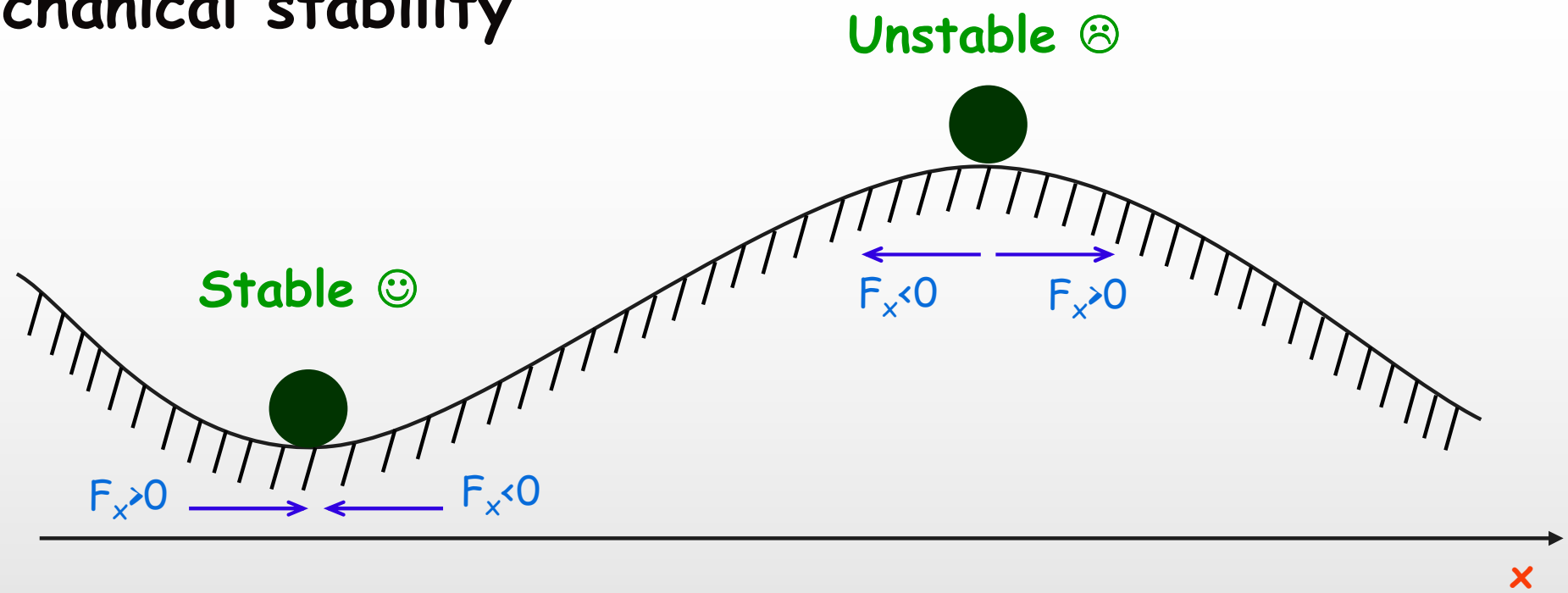
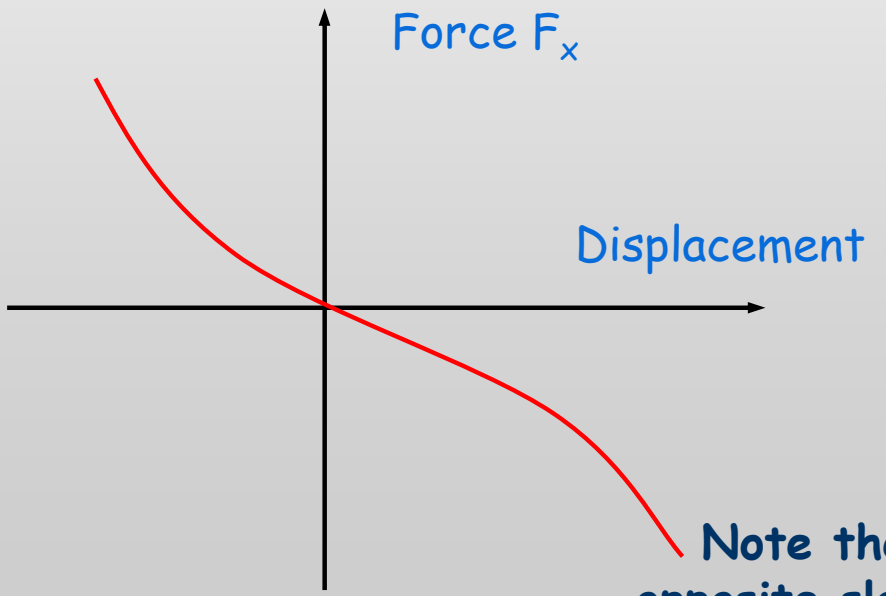
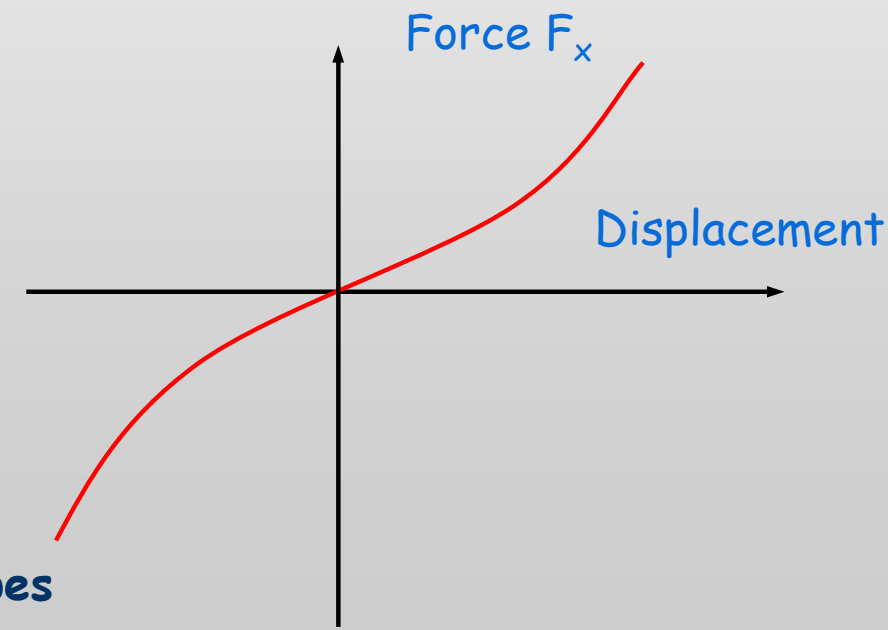
Mechanical stability







Note the opposite slopes

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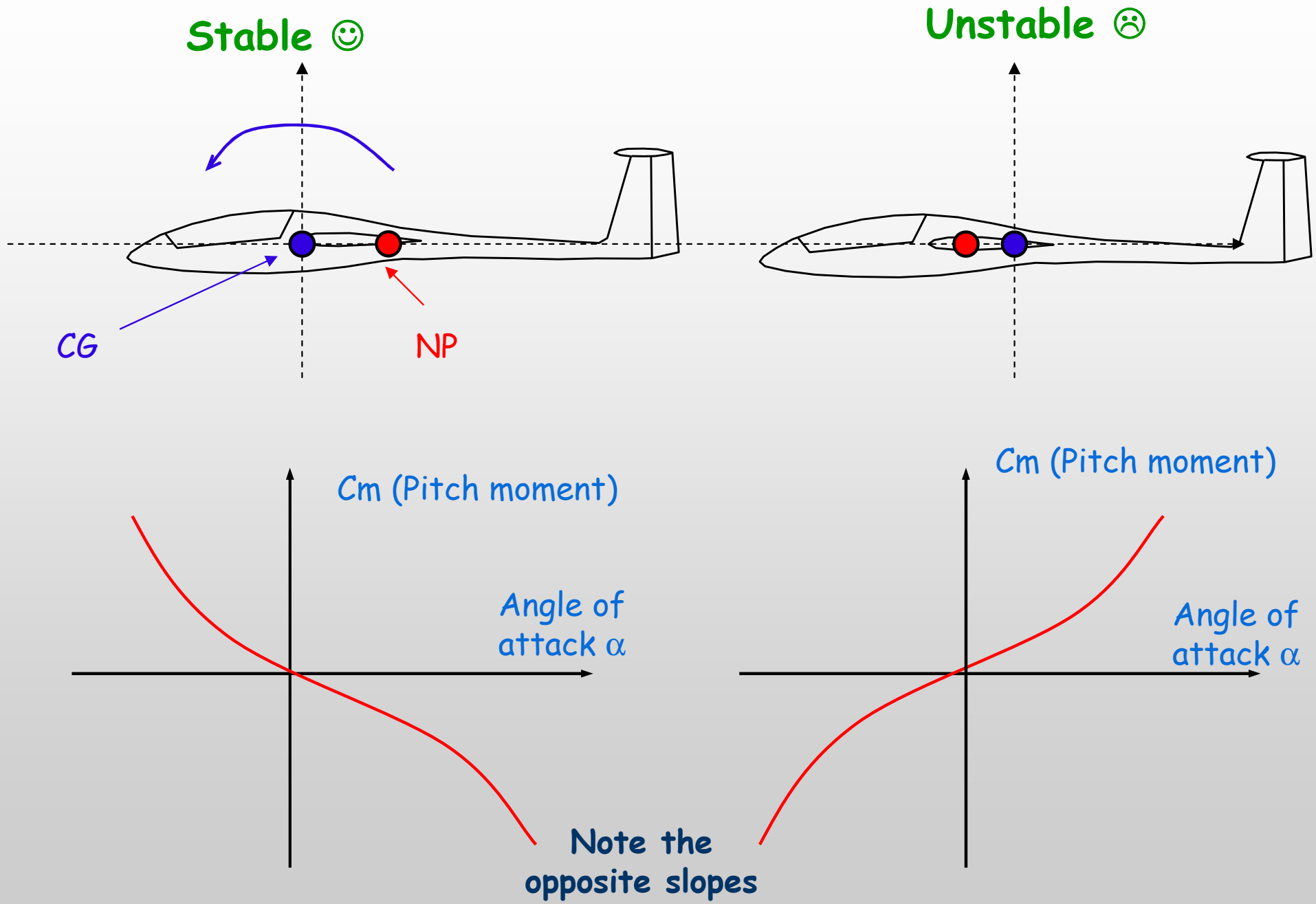
Aerodynamic stability



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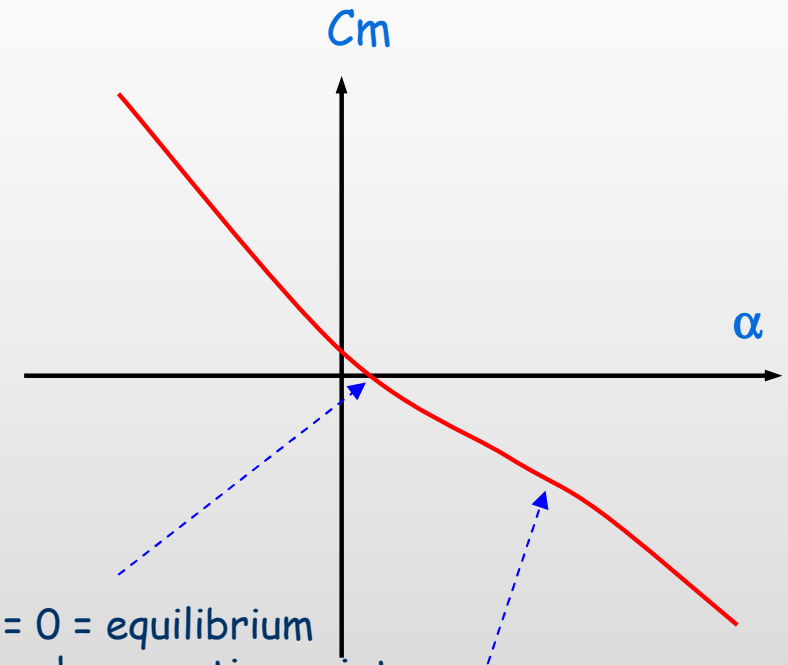
Understanding the polars Cm = f(α) and Cl = f(Cm)

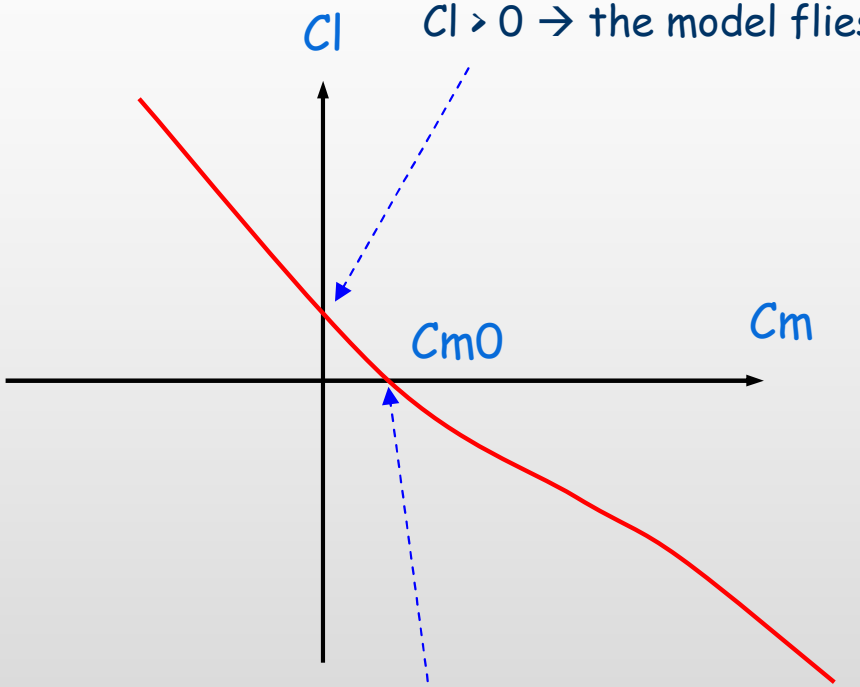
Note : Valid only for a whole plane or a flying wing



Cm = 0 = equilibrium
= the plane's operating point

Negative slope = Stability
The curve's slope is also the strength of the stabilizing force
→ **High slope = Stable sailplane !**

Cm = 0 → equilibrium
Cl > 0 → the model flies !



For information only :
Cm0 = Moment coefficient at zero-lift

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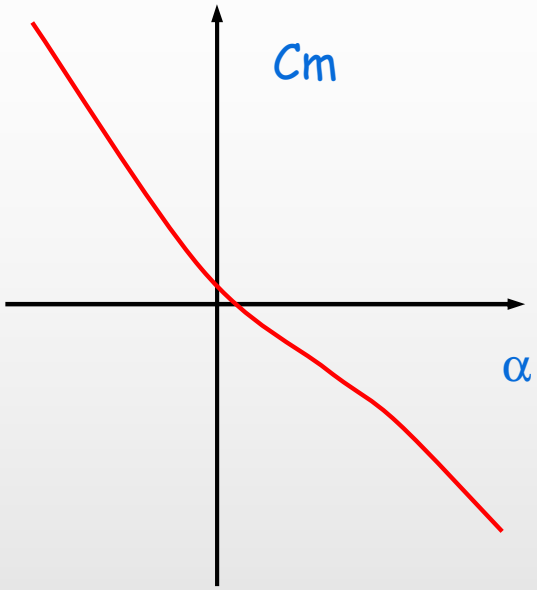
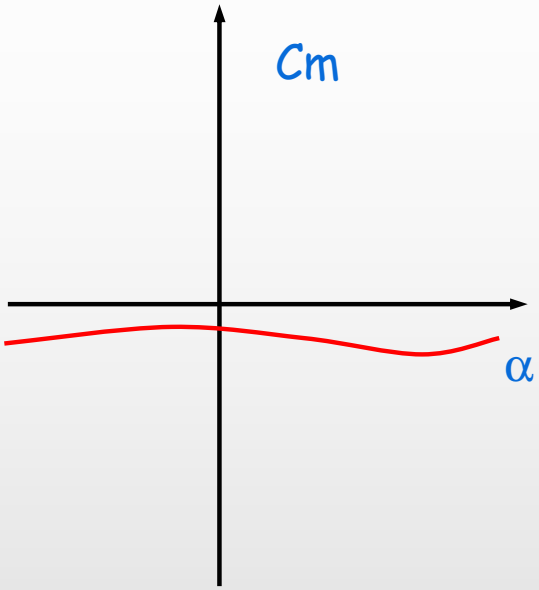
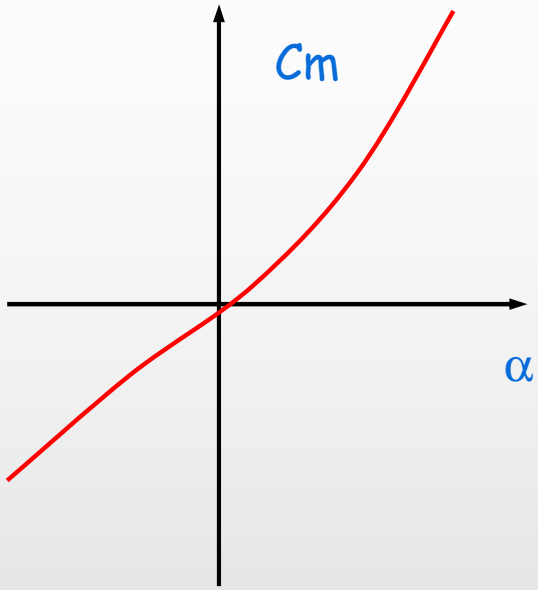
How to use XFLR5 to find the Neutral Point



Polar curve for $X_{CG} < X_{NP}$
The CG is forward of the NP
The plane is stable



Polar curve for $X_{CG} = X_{NP}$
Cm does not depend on α
The plane is unstable



Polar curve for $X_{CG} > X_{NP}$
The CG is behind the NP
The plane is stable…
The wrong way

By trial and error, find the X_{CG} value which gives the middle curve

For this value, $X_{NP} = X_{CG}$

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The tail volume (1) : a condition for stability ?

First the definition

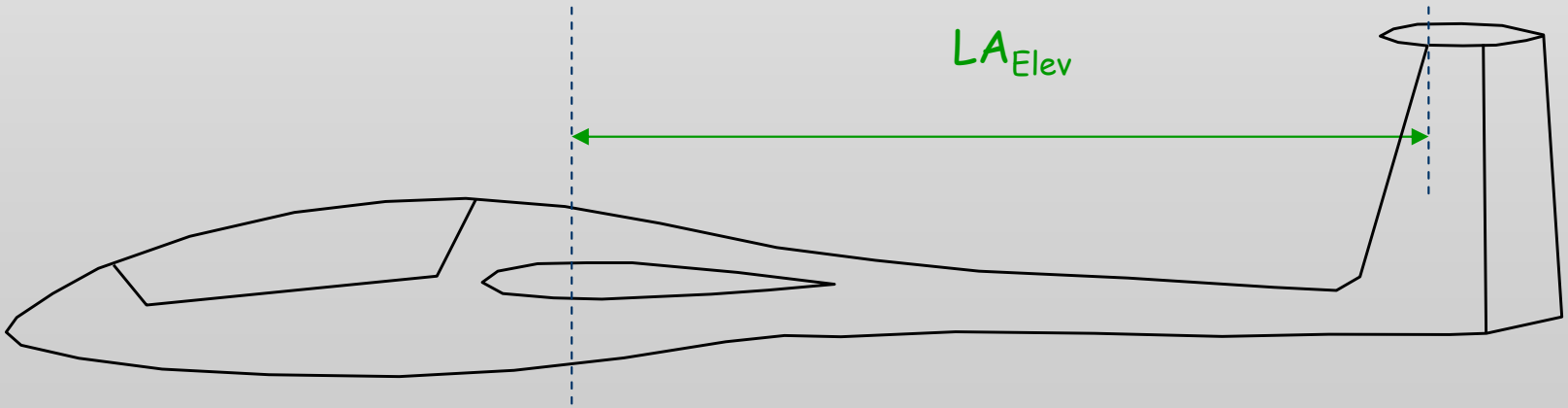
$$TV = \frac{LA_{Elev} \times Area_{Elev}}{MAC_{Wing} \times Area_{Wing}}$$

LA_{Elev} : The elevator's Lever Arm measured at the wing's and elevator's quarter chord point

MAC : The main wing's Mean Aerodynamic Chord

$Area_{Wing}$: The main wing's area

$Area_{Elev}$: The elevator's area



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Tail Volume (2)

Let's write the balance of moments at the wing's quarter chord point, ignoring the elevator's self-pitching moment

$$M_{Wing} + LA_{Elev} \times Lift_{Elev} = 0$$

M_{Wing} is the wing's pitching moment around its root $\frac{1}{4}$ chord point

We develop the formula using Cl and Cm coefficients :

$$q \times Area_{Wing} \times MAC_{Wing}\, Cm_{Wing} = - LA_{Elev} \times q \times Area_{Elev} \times Cl_{Elev}$$

where q is the dynamic pressure.

Thus :

$$Cm_{Wing} = -\frac{LA_{Elev} \times Area_{Elev}}{MAC_{Wing} \times Area_{Wing}} Cl_{Elev} = -TV \times Cl_{Elev}$$

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Tail Volume (3)

The elevator's influence increases with the lever arm

The elevator's influence increases with its area

$$Cm_{Wing} = -\frac{LA_{Elev} \times Area_{Elev}}{MAC_{Wing} \times Area_{Wing}} Cl_{Elev} = -TV \times Cl_{Elev}$$

The elevator has less influence as the main wing grows wider and as its surface increases

We understand now that the tail volume is a measure of the elevator's capacity to balance the wing's self pitching moment

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Tail Volume (4)

$$Cm_{Wing} = -TV \times Cl_{Elev}$$

- The formula above tells us only that the higher the TV, the greater the elevator's influence shall be
- It does not give us any clue about the plane's stability
- It tells us nothing on the values and on the signs of Cm and Cl
- Thus, an adequate value for the tail volume is not a condition sufficient for stability

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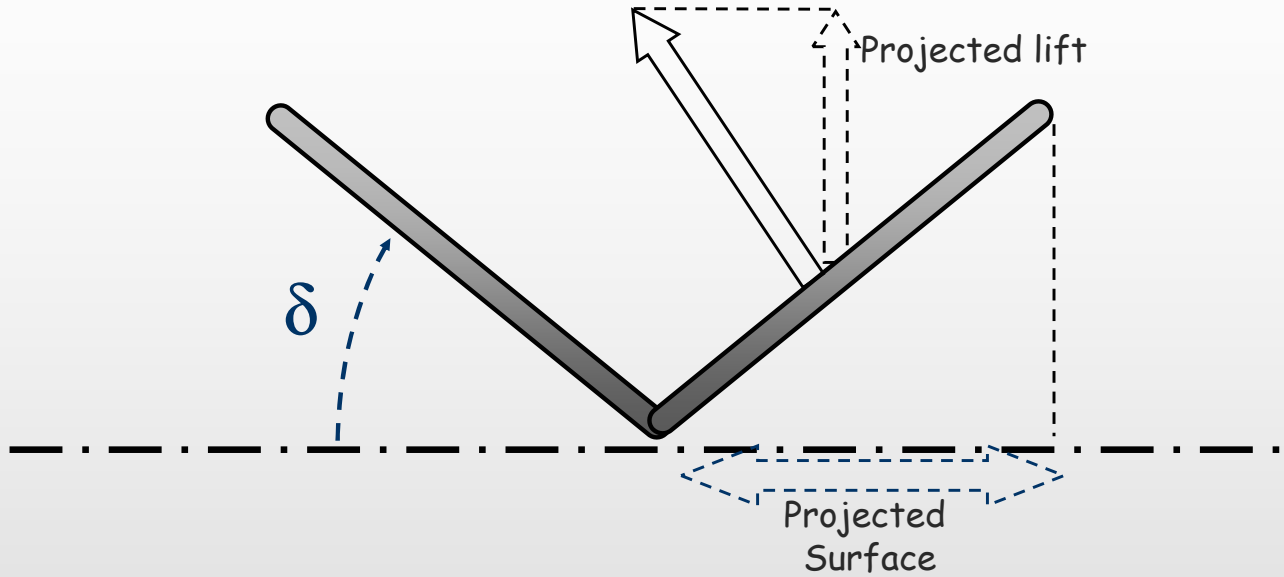
A little more complicated : V-tails

The method is borrowed from Master Drela (may the Aerodynamic Forces be with him)



The angle δ has a double influence:

1. It reduces the surface projected on the horizontal plane
2. It reduces the projection of the lift force on the vertical plane

… with a little math:

$$\text{Effective_area} = \text{Area}_{\text{Elev}} \times \cos^2\delta$$

$$TV = \frac{LA_{Elev} \times Area_{Elev} \times \cos^2\delta}{MAC_{Wing} \times Area_{Wing}}$$

Note : formula incorrect in XFLR5 v4.17 and QFLR5 v0.02: misses a conversion from degrees to radians
Corrected in QFLR5 v0.03

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The Static Margin : a useful concept

- **First the definition**

$$SM = \frac{X_{NP} - X_{CG}}{MAC_{Wing}}$$

- A positive static margin is synonym of stability
- The greater is the static margin, the more stable the sailplane will be
- We won't say here what levels of static margin are acceptable… too risky… plenty of publications on the matter also
- Each user should have his own design practices
- Knowing the NP position and the targeted SM, the CG position can be deduced…= X_{NP} - MAC x SM
- …without guarantee that this will correspond to a positive lift nor to optimized performances

How to use XFLR5 to position the CG

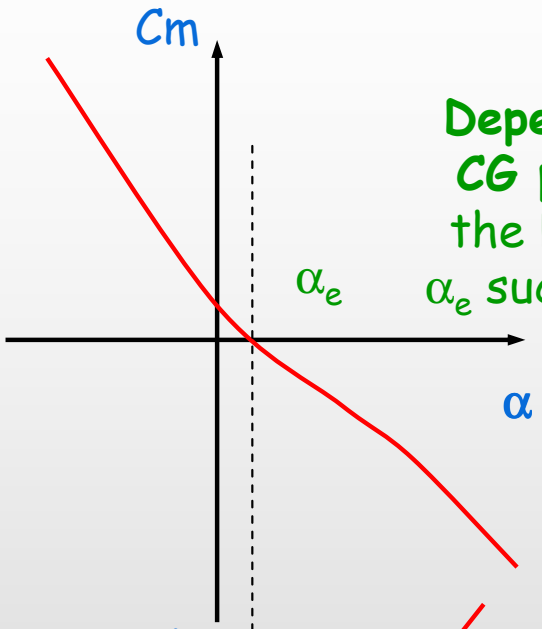
- **Idea N°1 : the most efficient**
  - Forget about XFLR5
  - Position the CG at 30-35% of the Mean Aero Chord
  - Try soft hand launches in an area with high grass
  - Move progressively the CG backwards until the plane glides normally
  - For a flying wing
    - Start at 15%
    - Set the ailerons up 5°-10°
    - Reduce progressively aileron angle and move the CG backwards
  - Finish off with the dive test

→ Works every time !

How to use XFLR5 to position the CG

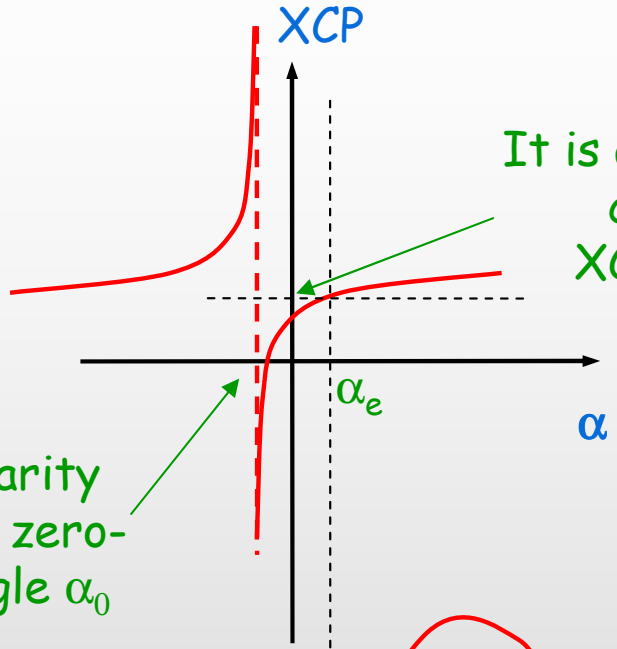
- **Idée N°2 : Trust the program**
  - Re-read carefully the disclaimer
  - Find the Neutral Point as explained earlier
  - Move the CG forward from the NP…
  - … to achieve a slope of Cm = f(α) comparable to that of a model which flies to your satisfaction, or
  - … to achieve an acceptable static margin
  - Go back to Idea N°1 and perform a few hand launches

Summarizing on the 4-graph view of XFLR5



①

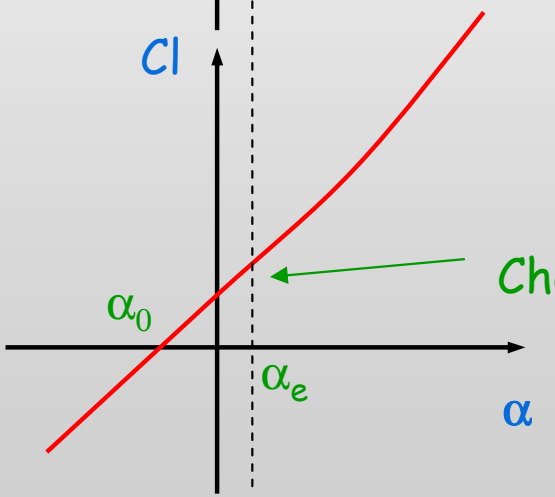
Depending on the CG position, get the balance angle α_e such that Cm = 0



③

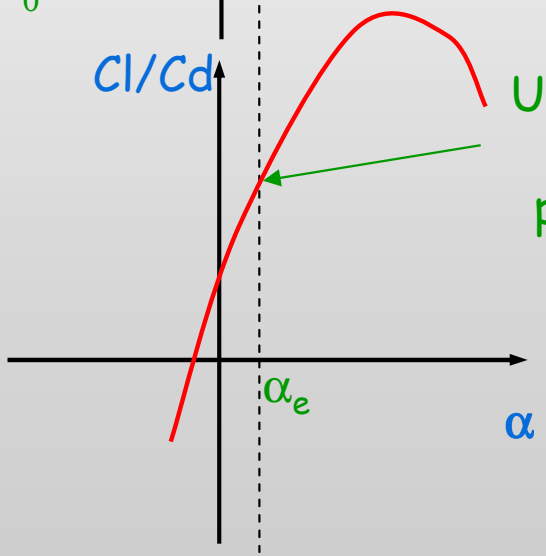
It is also possible to check that XCP =XCmRef for $\alpha = \alpha_e$

Singularity for the zero-lift angle α_0



②

Check that Cl>0 for $\alpha = \alpha_e$



④

Unfortunately, no reason for the performance to be optimal

Iterations are required to find the best compromise

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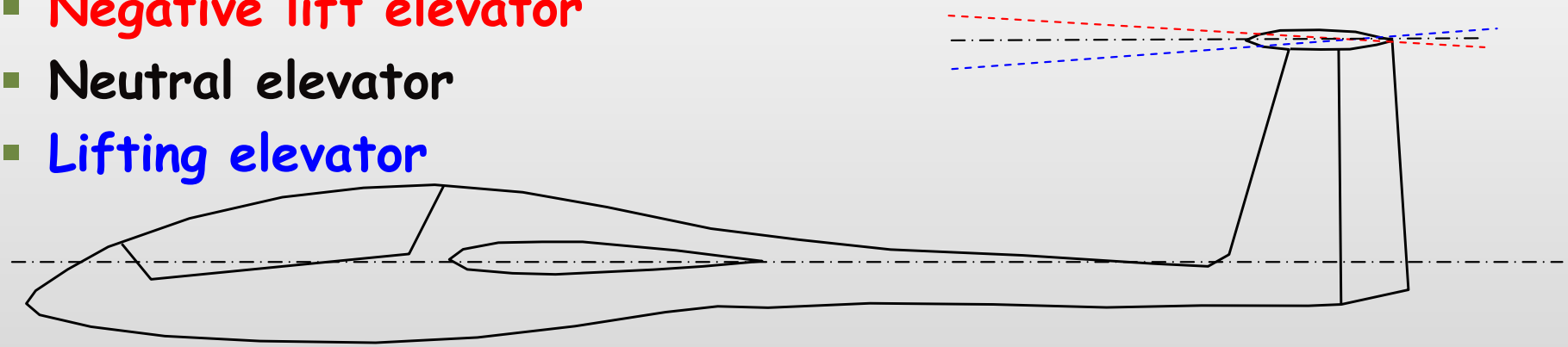
Consequences of the incidence angle

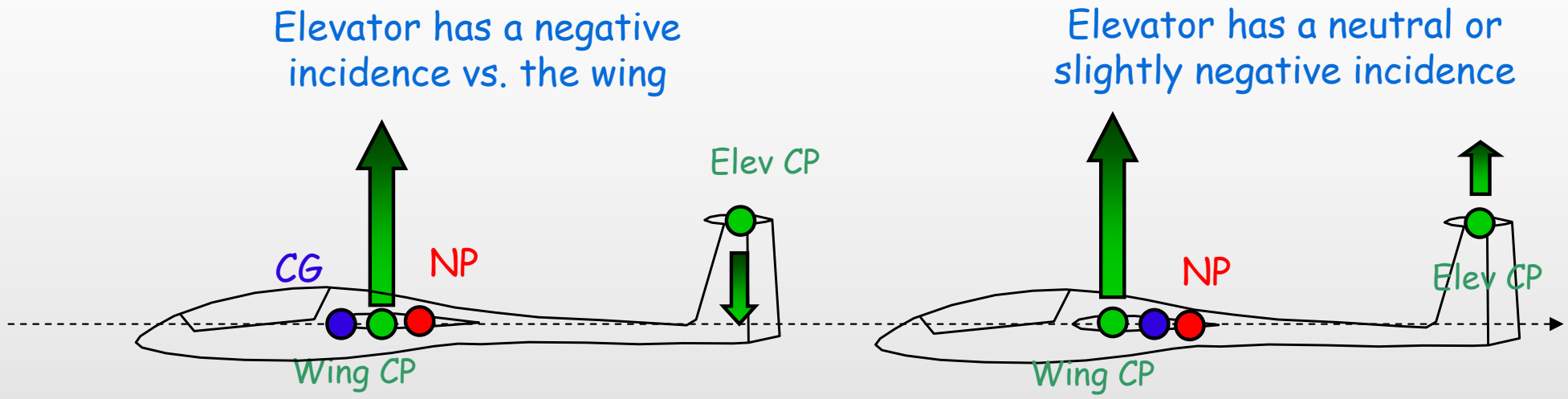
- To achieve lift, the wing must have an angle of attack greater than its zero-lift angle
- This angle of attack is achieved by the balance of wing and elevator lift moments about the CG
- Three cases are possible
  - Negative lift elevator
  - Neutral elevator
  - Lifting elevator



- Each case leads to a different balanced angle of attack
- For French speakers, read Matthieu's great article on http://pierre.rondel.free.fr/Centrage_equilibrage_stabilite.pdf

Elevator Incidence and CG position

- The elevator may have a positive or negative lift



- Both configurations are possible
- The CG will be forward of the wing's CP for an elevator with negative lift
- "Within the acceptable range of CG position, the glide ratio does not change much" (Matthieu Scherrer 2006)

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The case of Flying Wings

- No elevator
- The main wing must achieve its own stability
- Two options
  - Self stable foils
  - Negative washout at the wing tip

Self-Stable Foils

- The notion is confusing : The concept covers those foils which make a wing self-stable, without the help of a stabilizer
- Theory and analysis tell us that a foil's Neutral Point is at distance from the leading edge = 25% x chord
- But then… all foils are self-stable ??? All that is required is to position the CG forward of the NP
- What's the difference between a so-called self-stable foil and all of the others ???

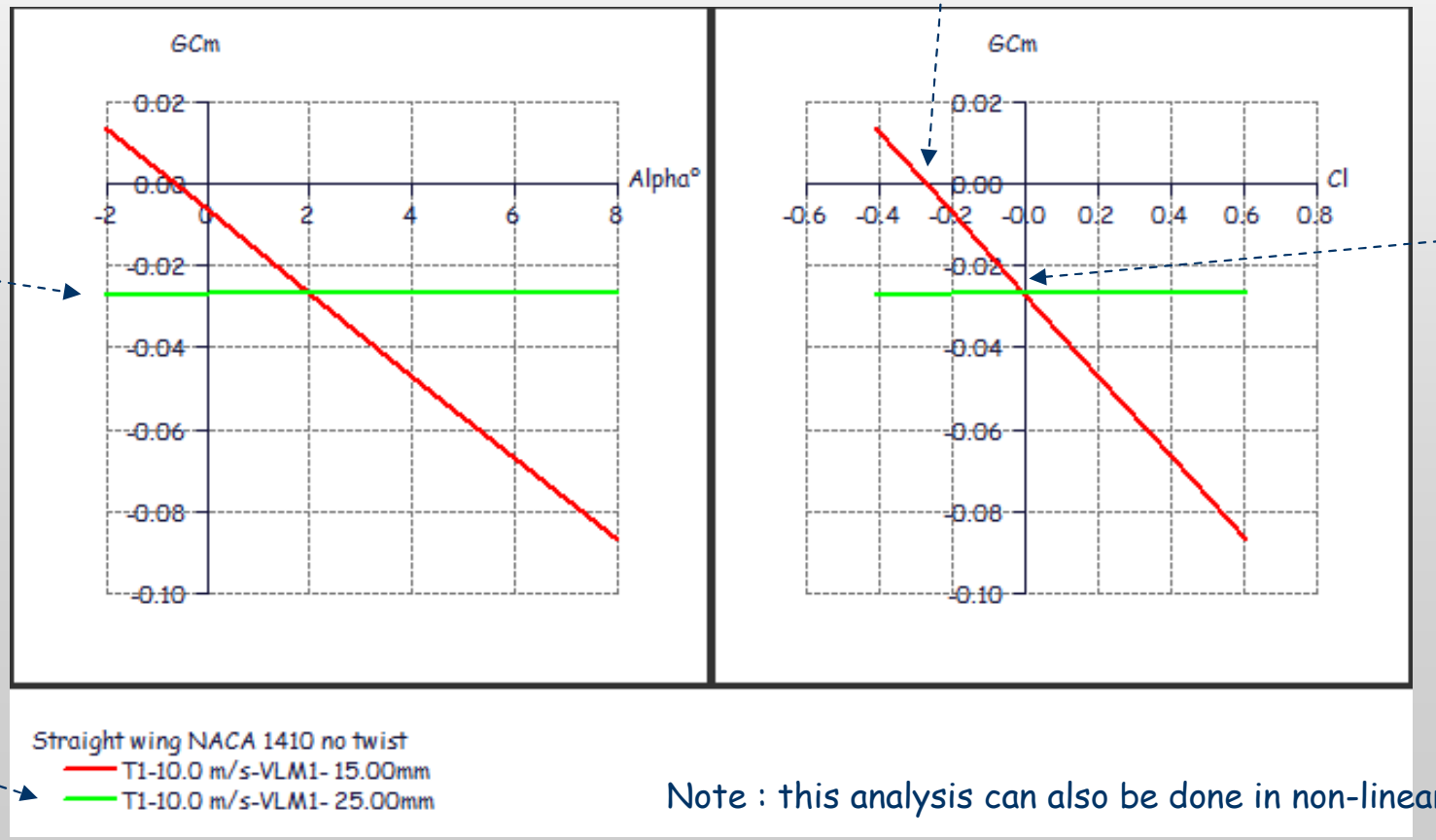
→ Let's explore it with the help of XFLR5

A classic foil

Consider a rectangular wing with uniform chord =100 mm, with a NACA 1410 foil reputedly not self-stable

Calculations confirm that the NP is at 25% of the chord

Unfortunately, at zero pitching moment, the lift is negative, the wing does not fly. That's the problem…



It is usually said of these airfoils that their zero-lift moment coefficient is negative
$Cm0 < 0$

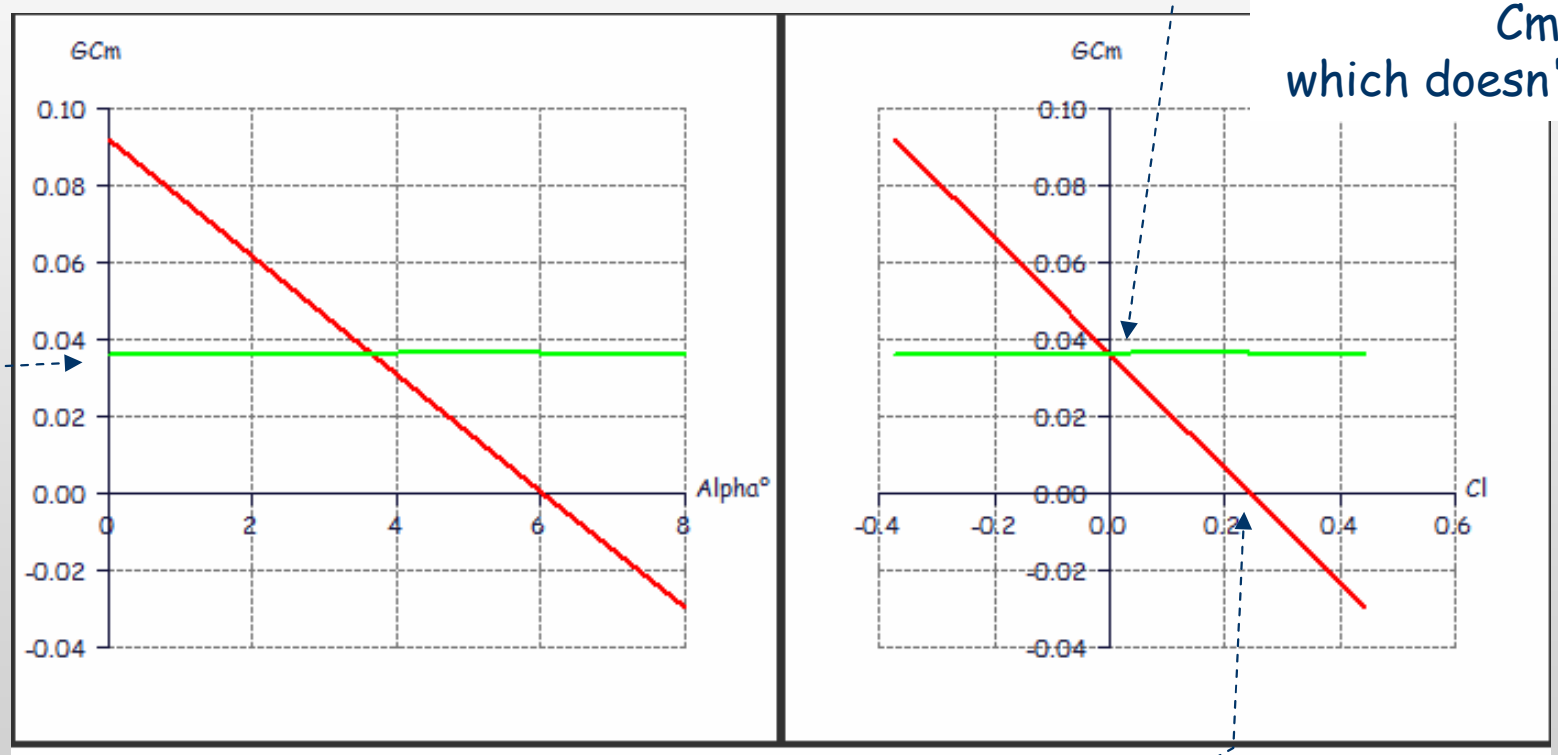
Note : this analysis can also be done in non-linear conditions with XFoil

A self-stable foil

Consider the same rectangular wing with chord 100mm, with an Eppler 186 foil known to be self-stable

The NP is still at 25% of the chord

It is usually said of these airfoils that "the zero-lift moment is positive", $Cm_0 > 0$ which doesn't tell us much



Straight wing Eppler 186 no twist
T1-10.0 m/s-VLM1- 10.00mm
T1-10.0 m/s-VLM1- 25.00mm

It would be more intuitive to say "the zero-moment lift is positive" : $Cl_0 > 0$, the wing flies!

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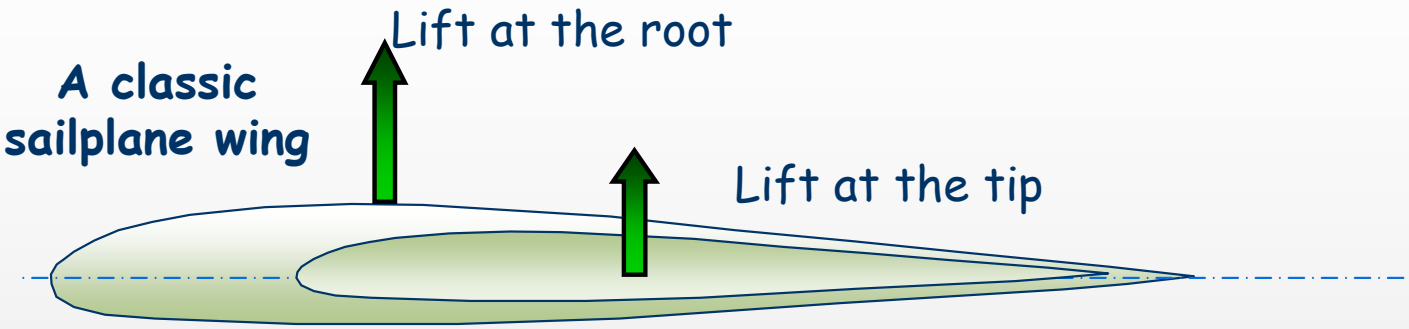
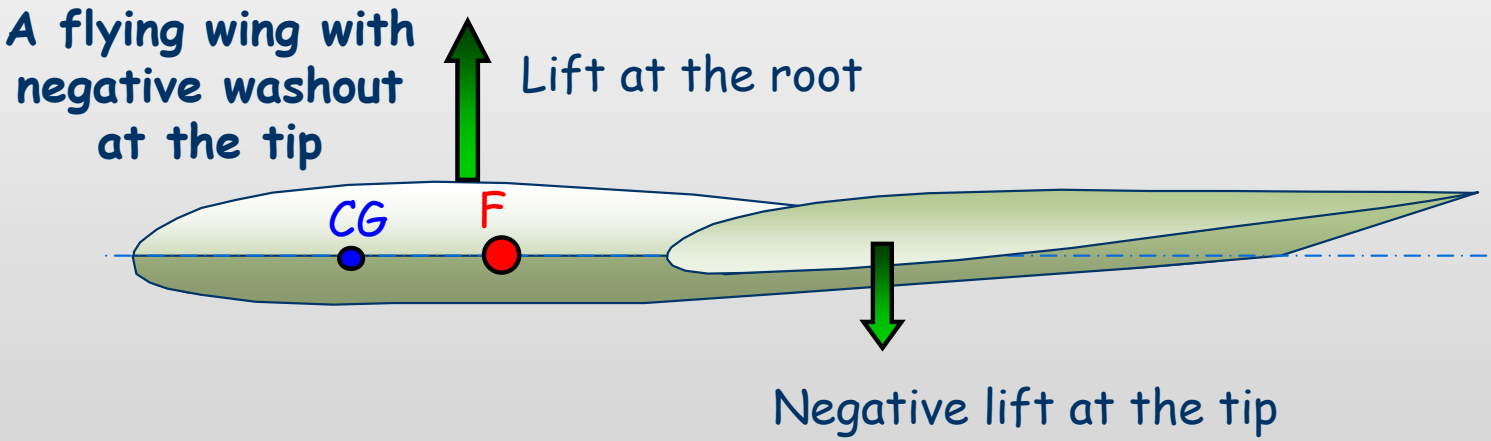
A more modern way to create a self-stable wing





The positive moment at the tip balances the negative moment at the wing's root

- The consequence of the negative lift at the tip is that the total lift will be less than with the classic wing
- Let's check all this with XFLR5

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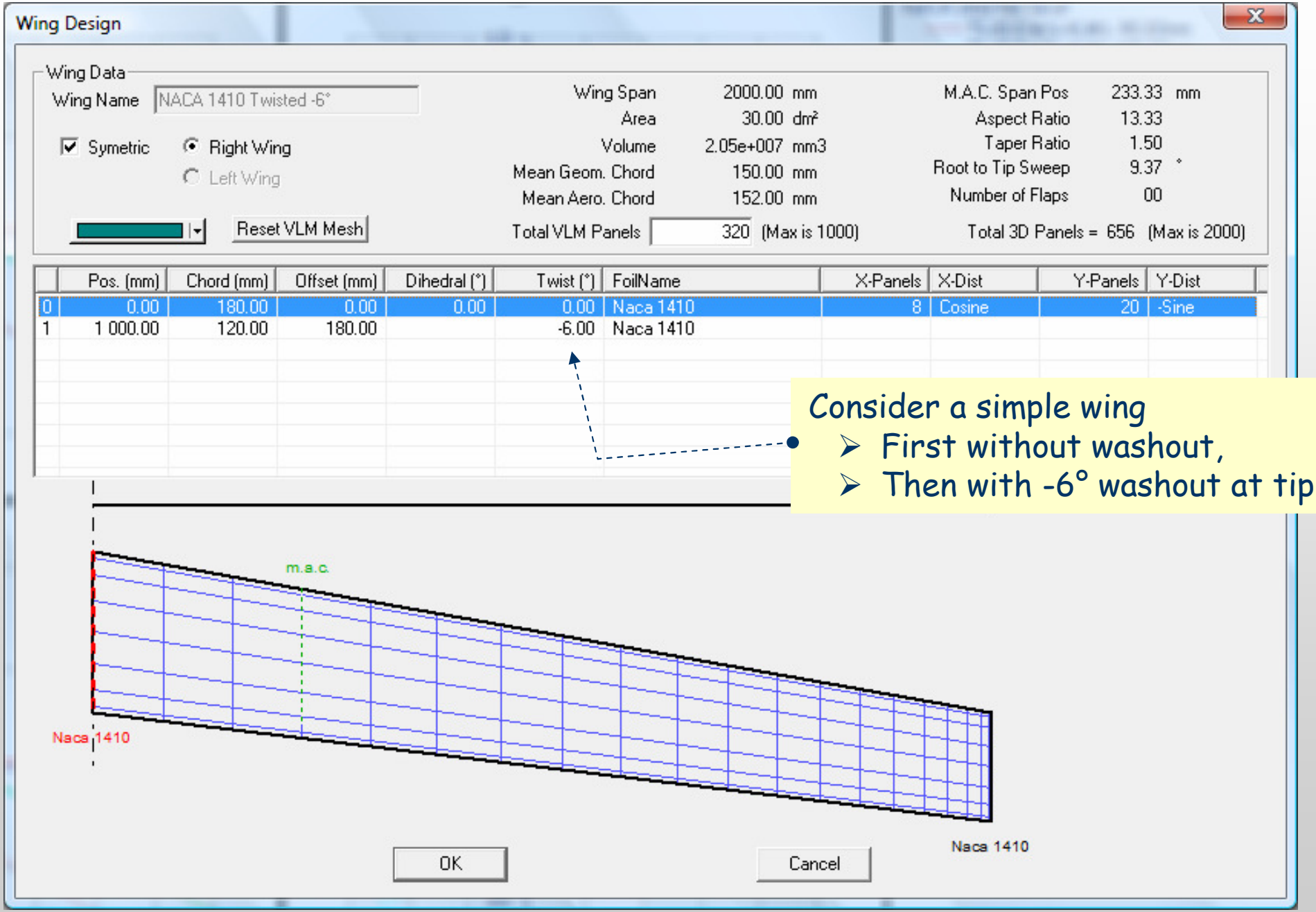
Model data

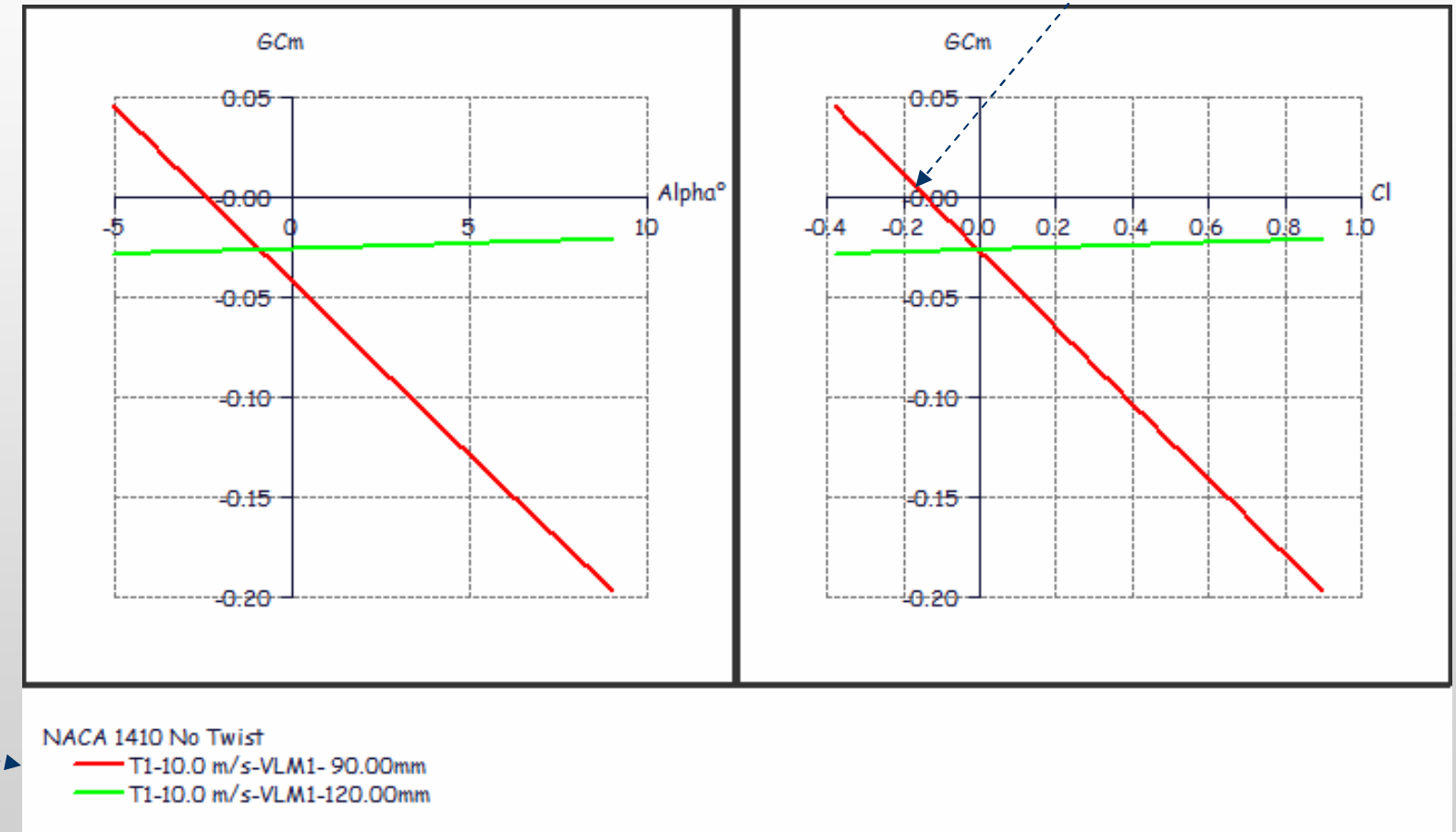


Consider a simple wing

- First without washout,
- Then with -6° washout at tip

Wing without washout

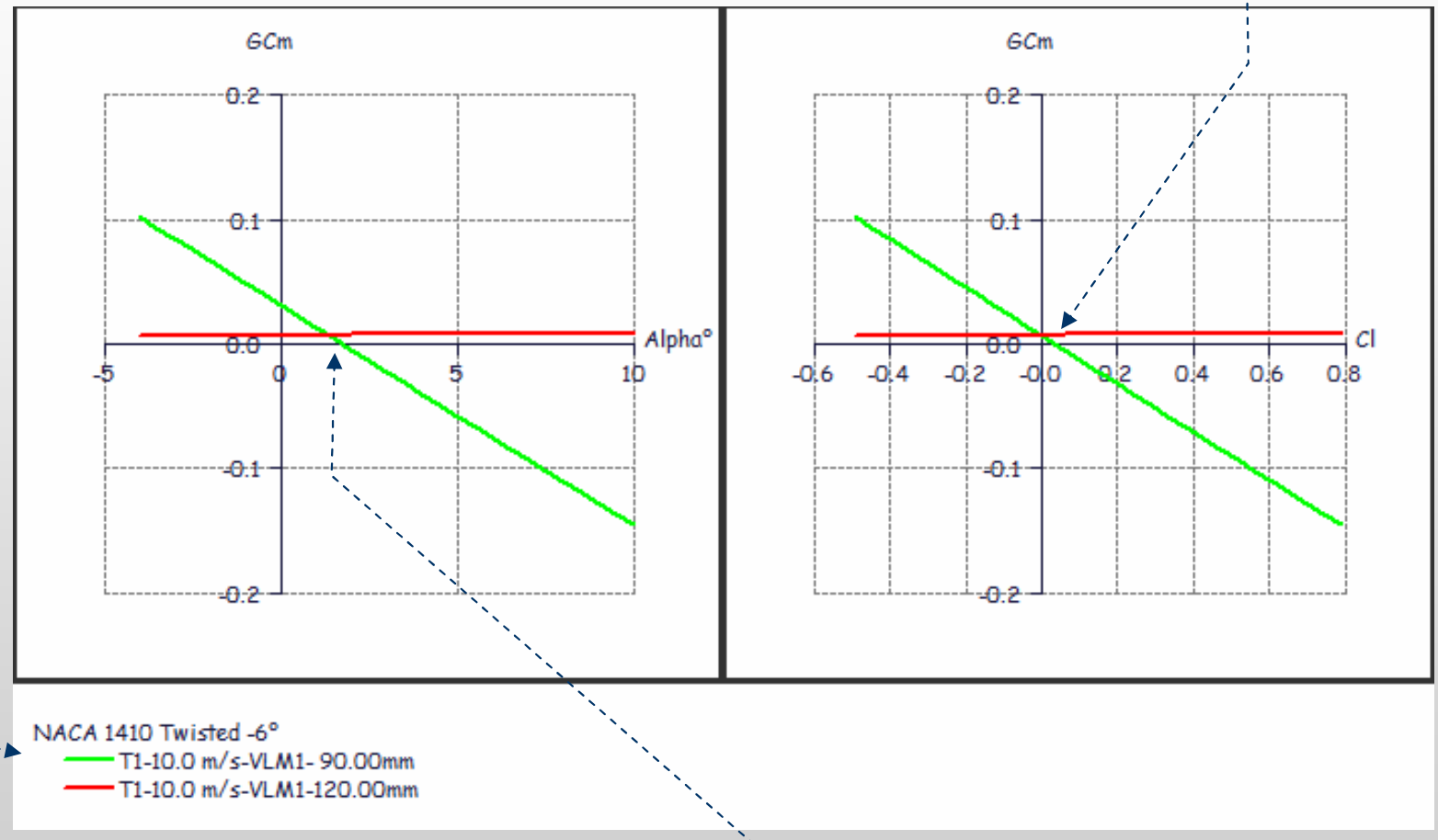
Unfortunately, at zero pitching moment, the lift is negative (Cl<0) : the wing does not fly



Consider a static margin = 10%

Wing with washout

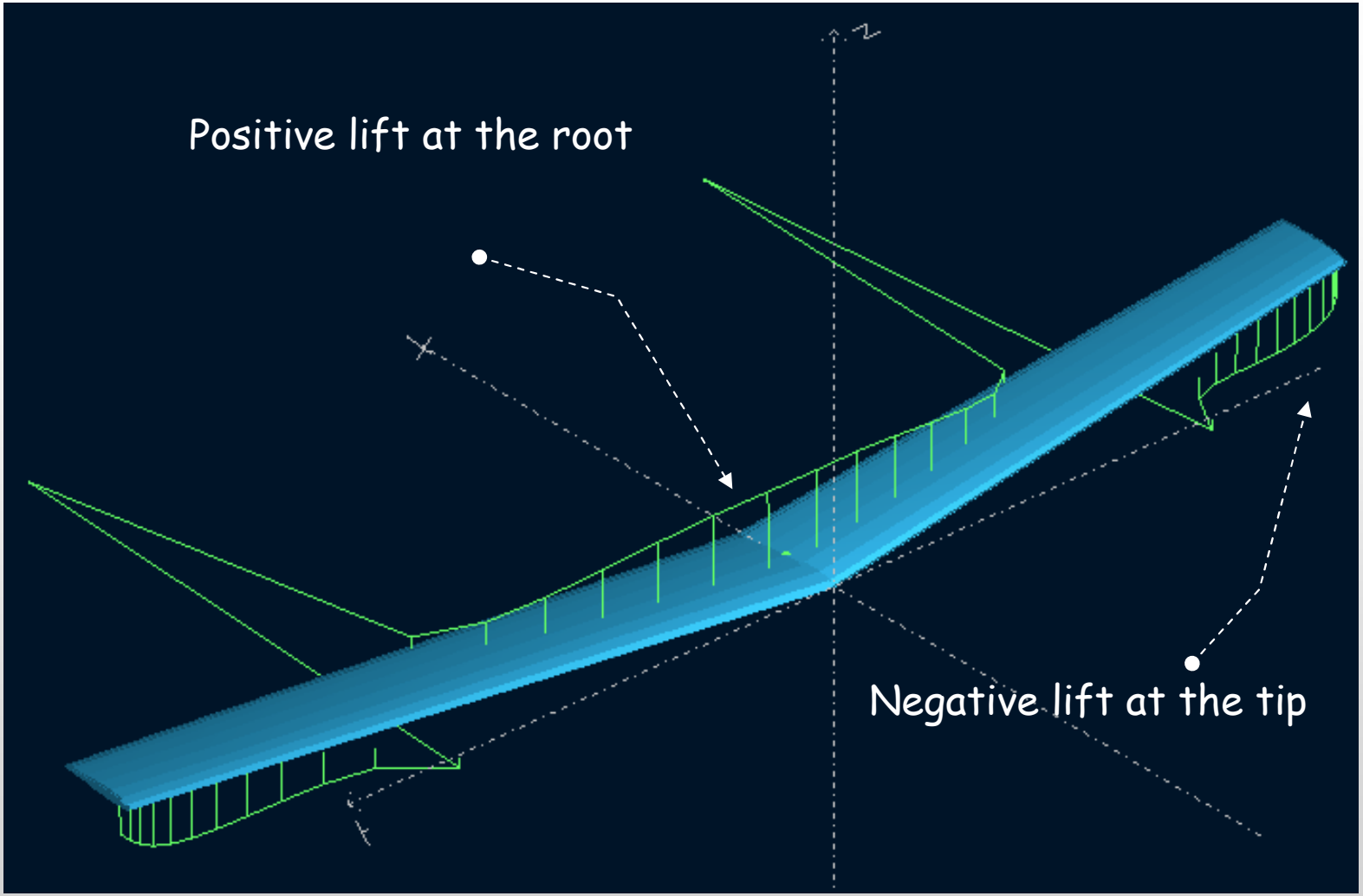
At zero pitching moment, the lift is slightly positive :
It flies !



Consider a static margin = 10%

Let's visualize in the next slide the shape of the lift for the balanced a.o.a α_e=1.7°

Lift at the balanced a.o.a



Part of the wing lifts the wrong way : a flying wing exhibits low lift

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About the Dive Test

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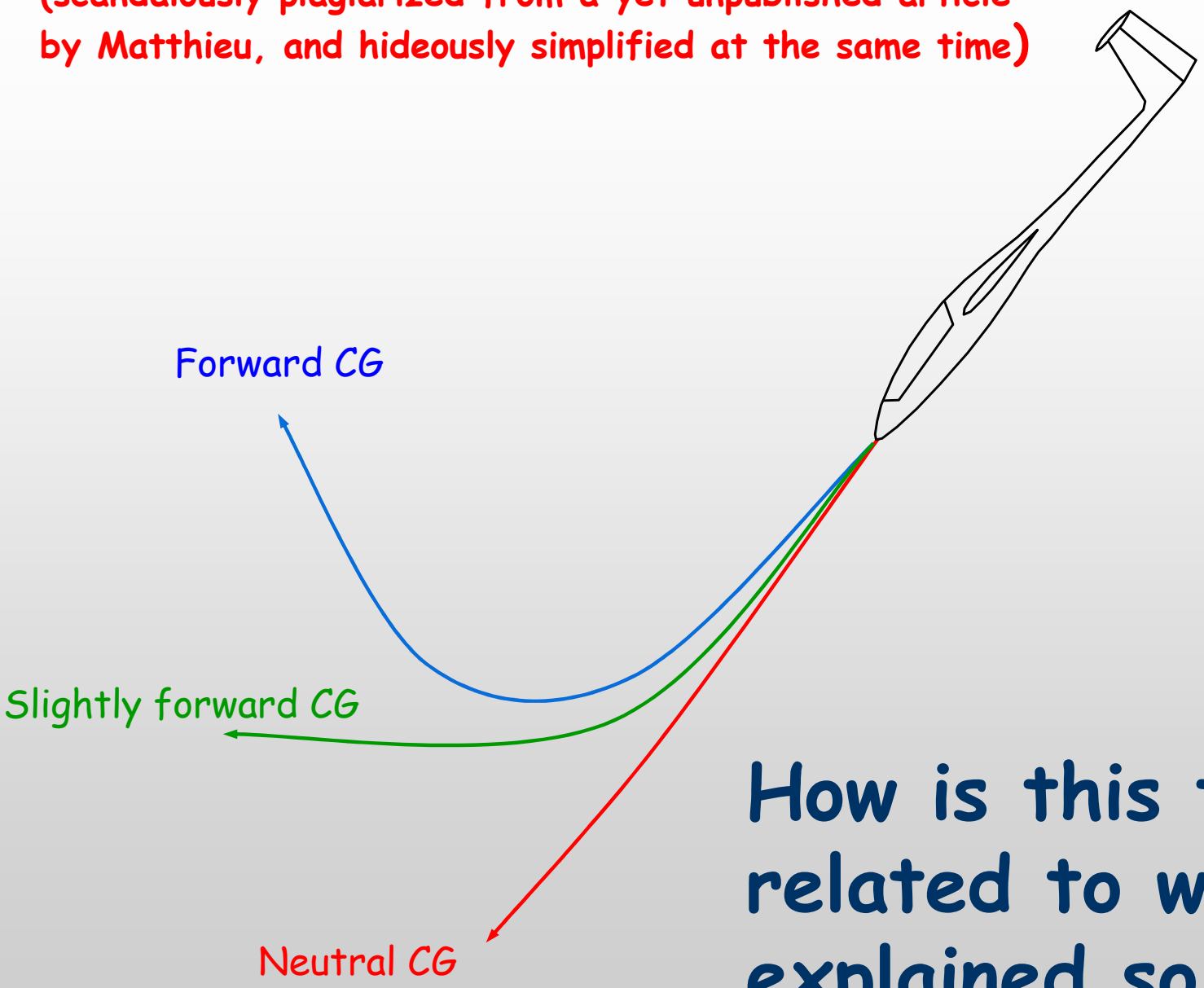
About the dive test

(scandalously plagiarized from a yet unpublished article by Matthieu, and hideously simplified at the same time)



How is this test related to what's been explained so far?

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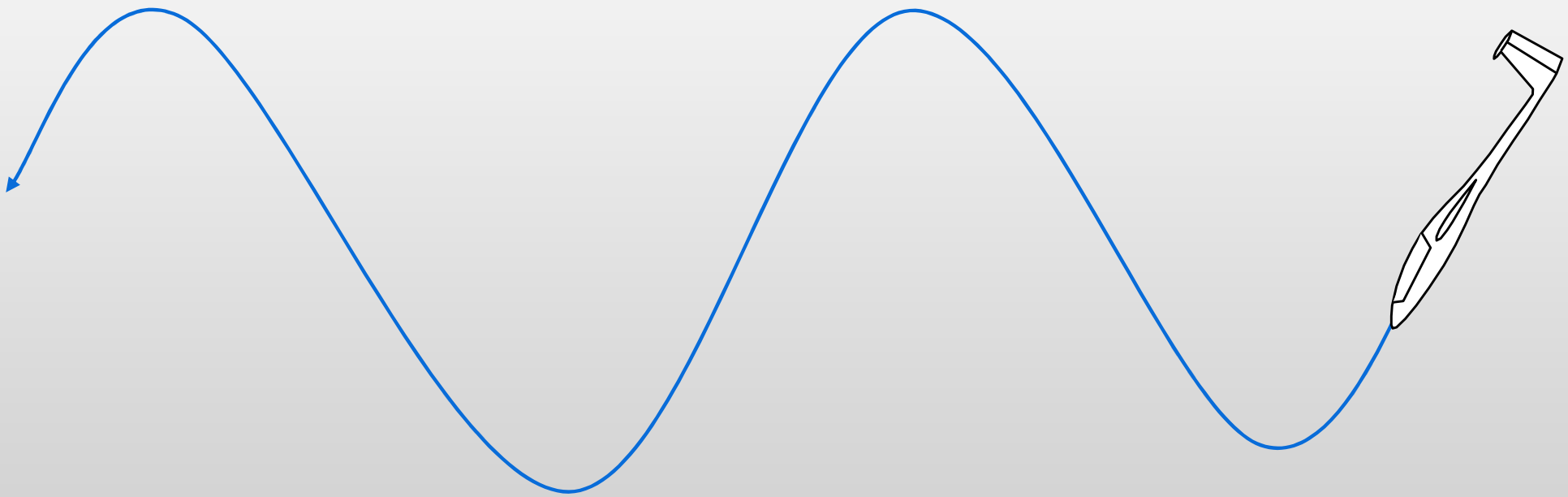
CG positioned forward

- If the CG is positioned forward enough, the sailplane will enter a phugoid after a pitching moment perturbation such as a dive



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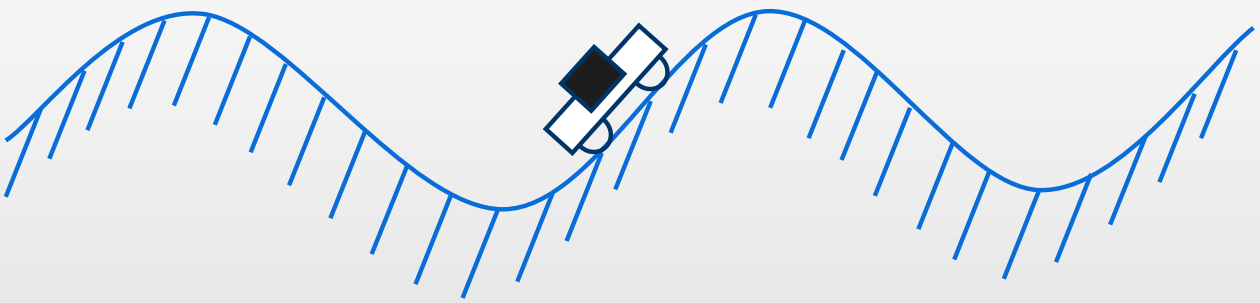
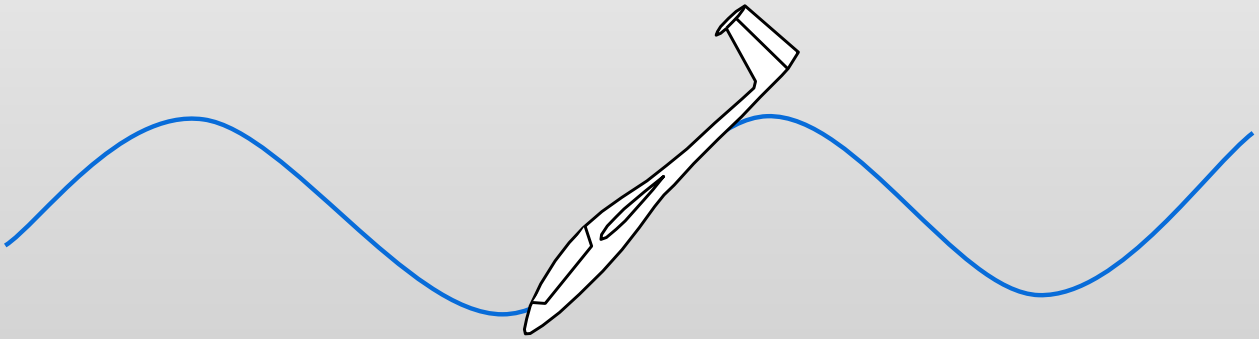
The phugoïd

… is a macroscopic mode of exchange between the Kinetic and Potential energies



Russian Mountains : Exchange is made by the contact force



Aerodynamic : Exchange is made by the lift force

This is not a harmonic oscillation but something else : the phugoid is not a sinusoid

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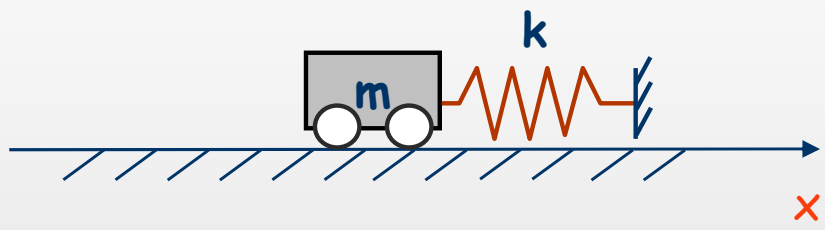
Stick to the phugoid

- **As the plane moves along the phugoid, the apparent wind changes direction**
- **From the plane's point of view, it's a perturbation**
- **The plane can react and reorient itself along the trajectory direction, providing**
  - That the slope of the curve $Cm = f(\alpha)$ is stiff enough
  - That it doesn't have too much pitching inertia
- **These two parameters are measured jointly by a natural frequency : "short period mode"**

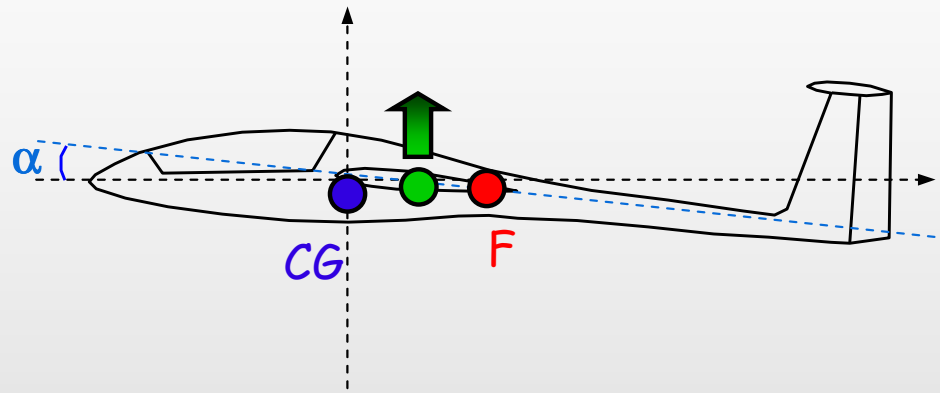
The "Short Period Mode"

...is a harmonic type oscillation (i.e. sinusoidal)

Mechanical Oscillation : the spring stiffness is the callback force



Aerodynamic Oscillation : the callback force is the slope of the curve Cm=f(α)



The mass may vibrate at the end of the spring. If so, it will be at a natural frequency solution of a simple differential equation.

$$m\frac{d^2x}{dt^2} + kx = 0$$

$$\text{Frequency} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

Similarly

$$J\frac{d^2\alpha}{dt^2} + K\alpha = 0$$

$$\text{Frequency} = \frac{1}{2\pi}\sqrt{\frac{K}{J}}$$

J is the inertia w.r.t the y-axis (pitching)

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The pitching moment "Short Period Mode"

$$\text{Frequency} = \frac{1}{2\pi}\sqrt{\frac{K}{J}}$$

1. If the frequency is close to that of the phugoid, the modes will interact : no good !
2. The coefficient K can be evaluated with XFLR5 :

$$K = \frac{1}{2}\rho V^2 \, S_{wing} \, M.A.C. \frac{dC_m}{d\alpha}$$

Negative slope of the curve
Cm = f(α)

3. The inertia J is difficult to evaluate without a CAD program …
4. …however, if it is evaluated, then a tool like AVL can provide the evaluation of the frequency, and much more !

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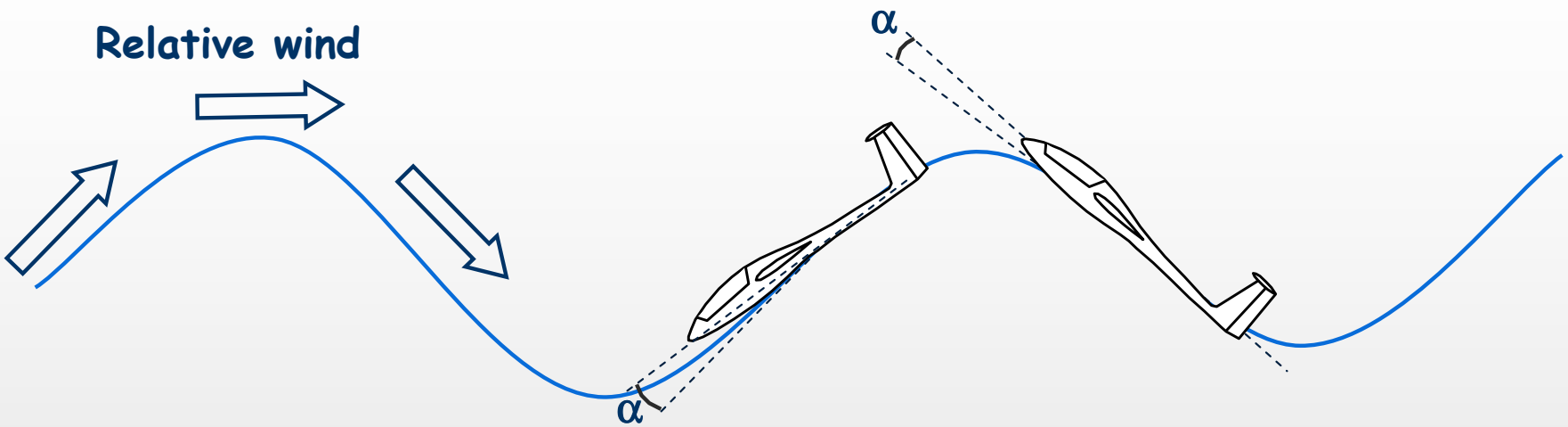
The mechanism of the phugoid



Dive → The sailplane accelerates → α Constant → Cl constant → → At iso-Cl, The lift increases as the square power of the speed $L = \frac{1}{2} \rho S V^2 Cl$ → The sailplane rises and slows down → The lift decreases → Dive

The difficulty is in the restriction "α constant"
If that's not the case, the phugoid disappears

Explanation

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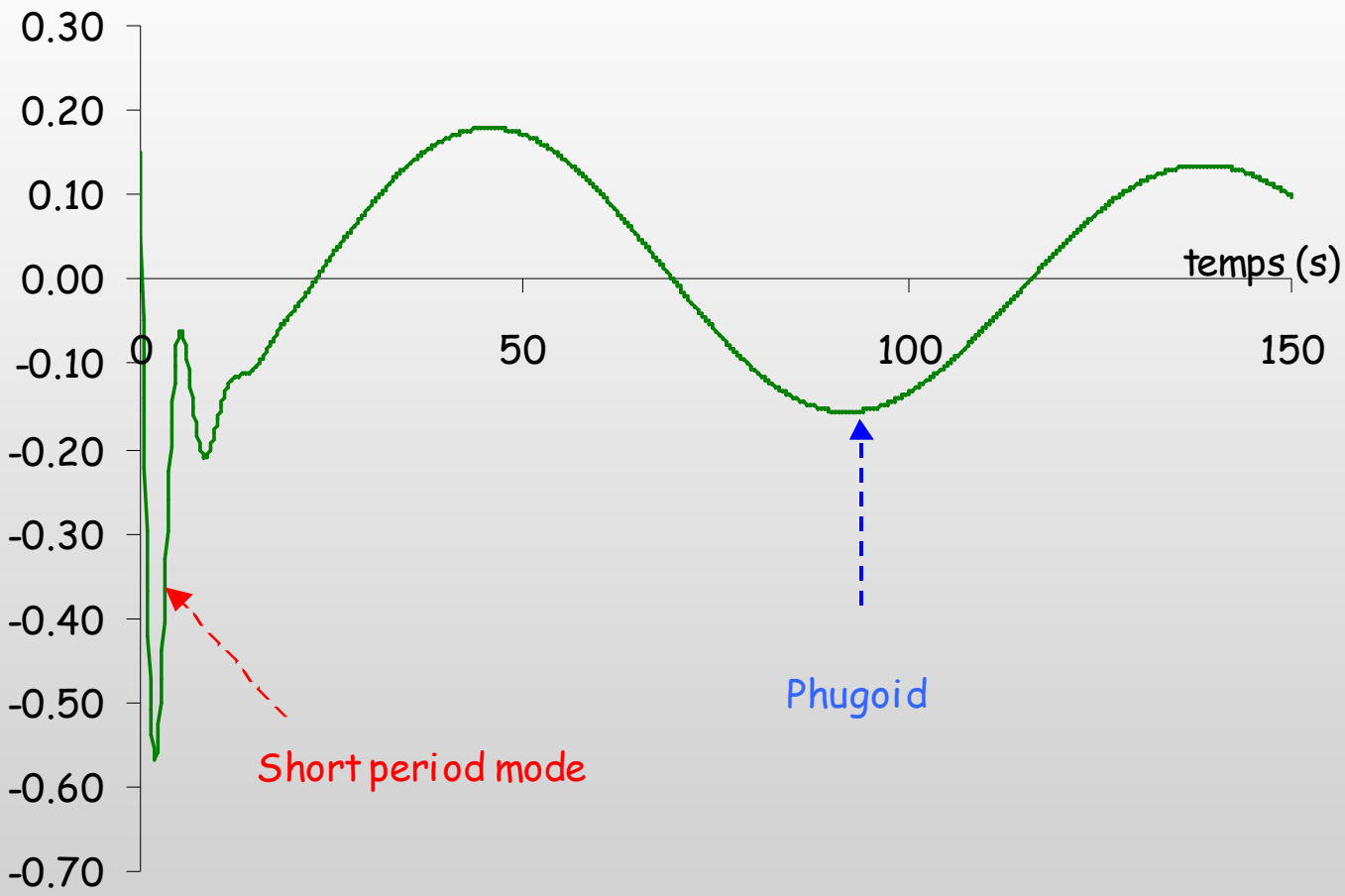
Example of de-coupled modes

- Pitching response of a real scale jet when disturbed by a pitching perturbation



- The period of the "short period mode" is a few seconds, whilst the period for the phugoid is something like a 100 seconds
- The two modes are de-coupled (and damped) making the plane safe

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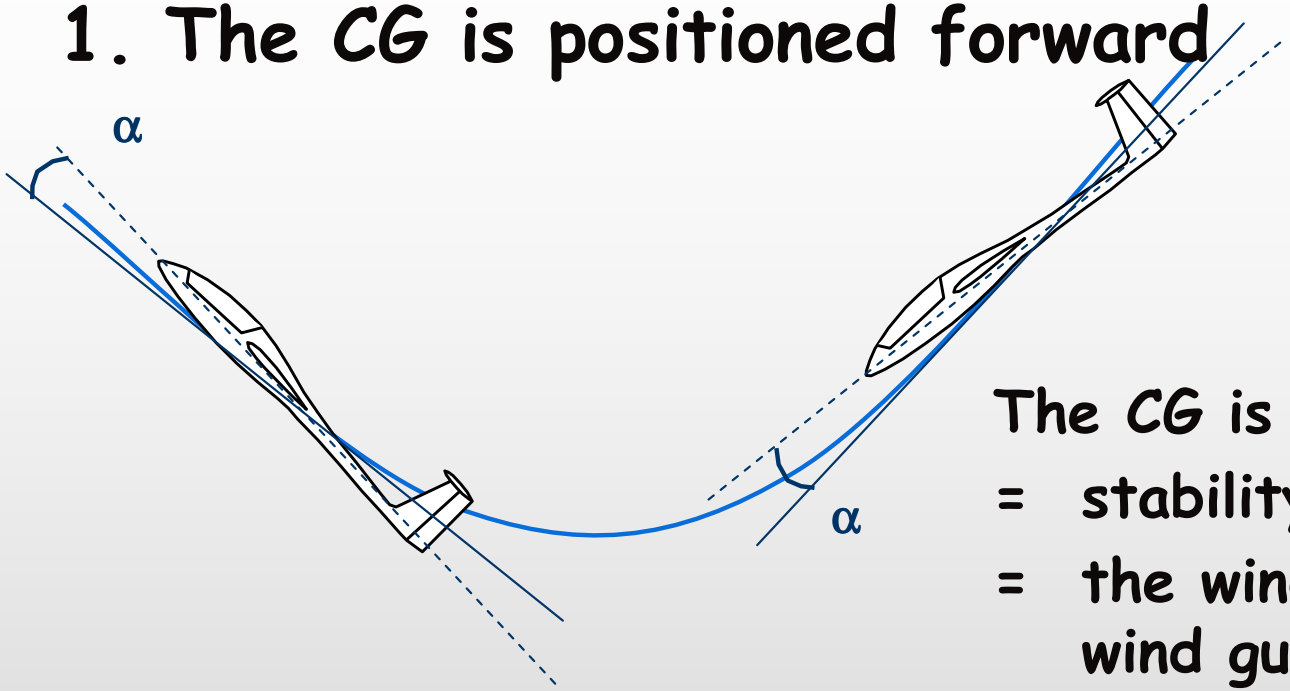
Summarizing :

1. The CG is positioned forward



The CG is positioned forward

= stability

= the wind vane which follows the wind gusts

- The two modes are coupled
- The relative wind changes direction along the phugoid…
- … but the plane maintains a constant incidence along the phugoid, just as the chariot remains tangent to the slope
- The sailplane enters the phugoid mode

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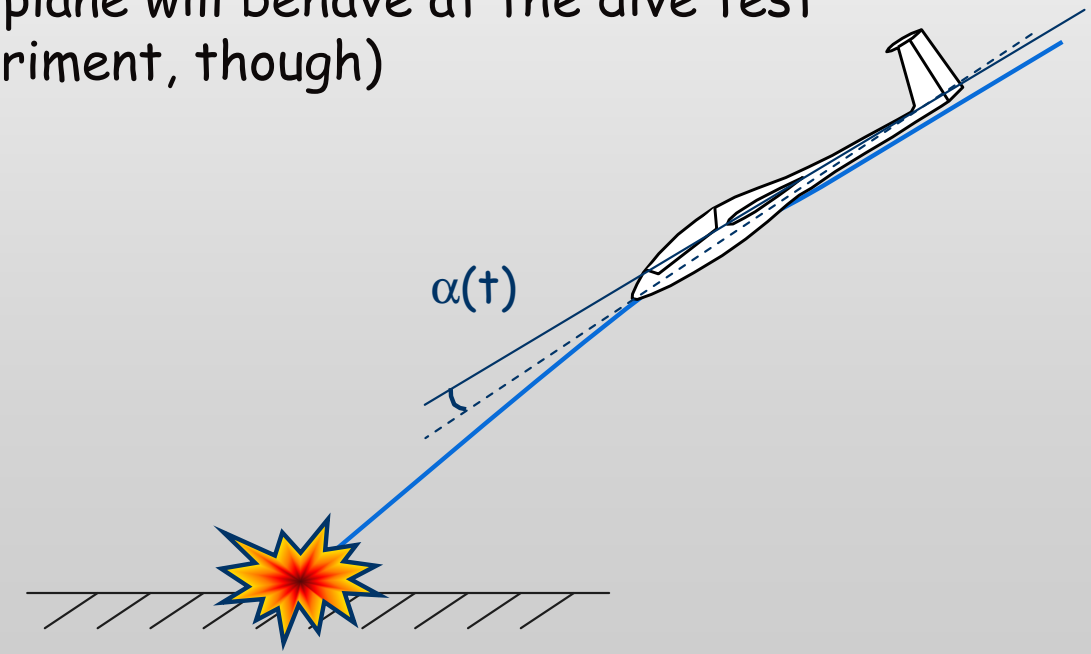
2. The CG is positioned aft

Remember that backward CG = instability = the wind vane which amplifies wind gusts

- The two modes are coupled
- The incidence oscillation $\alpha(t)$ amplifies the phugoid,
- The lift coefficient is not constant during the phugoid
- The former loop doesn't work any more
- The phugoid mode disappears
- No guessing how the sailplane will behave at the dive test (It's fairly easy to experiment, though)



Note : Dynamic instability is much more complex than this simplified description would suggest

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That's all for now

Good design and nice flights ☺

Needless to say, this presentation owes a lot to Matthieu Scherrer ; thanks Matt !

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